IAEA Coordinated Research Project on Development of Harmonized QA/QC Procedures for Maintenance and Repair of Nuclear Instruments

Test Procedure for Ionization Chamber Detectors

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1. OBJECTIVE AND SCOPE.

1.1 . Objective.

To establish a test procedure for the verification of performance, measurement of the basic electrical characteristics and the radiation response of ionization chamber detectors as used in direct current mode.

1.2. Scope

This procedure applies to ionization chambers vented to the atmosphere and pressurized chambers working in direct current mode as used in nuclear applications.

2. NOTATION AND DEFINITIONS.

2.1. Notation.

$\dot{X}$ Exposure rate.

2.2. Definitions.

2.2.1. Ionization chamber.
Radiation detector consisting of a chamber filled with air or gas, in which an electric field inside the detector is provided for the collection of charges associated with ions and electrons produced in the measuring volume of the detector by the ionizing radiation.

2.2.2. Dosimeter.
Equipment that uses an ionization chamber for measurements of air Kerma, absorbed dose, or their corresponding rates.

2.2.3. Electrometer.
Equipment used to measure very small electrical currents (in the order of 10^{-8} A to 10^{-15} A) or small electrical charges (in the order of 10^{-12} to 10^{-15} C).

2.2.4. Exposure rate ($\dot{X}$).
Exposure rate is the ratio between dX and dt, where dX is the differential of exposure in the time interval dt. The unit of exposure rate is $C.kg^{-1}.s^{-1}$.

2.2.5. Sensitivity.
The ration between the current produced by an ionization chamber and the exposure rate, given for a radiation source. The isotope employed must be specified.

2.2.6. Leakage current.
Any current which is not produced by radiation and could be added to the measured ionization current causing an unwanted component of the measured signal.

2.2.7. Error of measurement.
The difference remaining between the measured value of a quantity and the true value of that quantity.

2.2.8. Air Kerma.
Kinetic energy released in a unit of mass of air.

3. DEVELOPMENT.

A methodology is described to test the electrical characteristics and the radiation response of ionization chamber detectors. A flow chart is shown in the Annex I which summarizes this process.

3.1. Recommended equipment and performance requirements.

The list of all recommended equipment required for testing of ionization chamber detectors is shown below. Alternative equipment may be used as long as the substitute equipment has specifications at least as good as those listed. Additionally all equipment should be calibrated

3.1.1 Electrometer (for example electrometer Keithley, model 35617 or equivalent).

3.1.2 Radioactive source, for example a ^{90}Sr source with an activity of 3.7×10^7 Bq (1 mCi) or less. In some ionization chamber with thin window it is advisable to use a low activity Alfa source of 37000 Bq (1µCi) such as ^{241}Am.

3.1.3 Digital multimeter.

3.1.4 High voltage probe.

3.1.5 Chronometer.

3.2. Test Conditions

3.2.1 Background Radiation
Be sure that the only contribution to the detector counting is the natural background, avoiding the contribution due to any additional radioactive source.

3.2.2.- Temperature
The temperature is a factor that influences greatly the response of ionization chambers, the standard temperature for ionization chambers is 273.15 ºK, therefore in any measurement the actual temperature has to be recorded.

3.2.3.- Humidity
The humidity can modify the isolating characteristics between electrodes in the ionization chamber and the measured current could be influenced by additional

leakage currents, the relative humidity must be between 47 % and 53 %, in any case the actual humidity has to be recorded.

3.2.2.- Atmospheric Pressure

The atmospheric pressure affects the response of ionization chambers vented to the atmosphere, the standard pressure is 760 mm Hg (101.3 kPa), in any case the actual atmospheric pressure has to be recorded.

3.3. Ionization chamber.

3.3.1 Ionization current measurement.

An ionization chamber needs a bias voltage, as illustrated in figure 1. The voltage produces an electric field that helps to collect the charges inside the detector and these charges (electrons and ions) are proportional to the radiation. Typical ionization currents or ionization charges in most applications are extremely small (in the order of 10^{-9} to 10^{-15} A or 10^{-12} C to 10^{-15}C). These magnitudes are very small to be measured with an ordinary instrument. Thus, we need an especial instrument called electrometer. This instrument could measure the current or electric charge by two methods.

a) Current measurement.

An electrometer indirectly measures the current by sensing a voltage drop across a resistance (typically with a value between 10^{9} Ω to 10^{12} Ω) placed in the measuring circuit, as shown in Fig. 1.

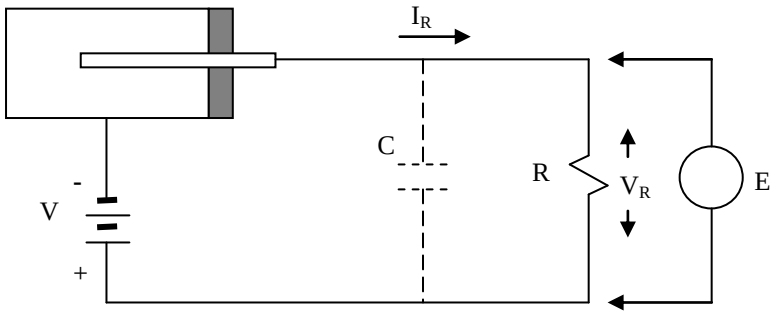


Fig. 1. Ionization current measurement.

Where,
E = Electrometer.
R = Input resistance of electrometer.
C = Capacitance of chamber.

I_R = Ionization current.

The voltage drop across the resistance (R) is

$$V_R = I_R.R \qquad (1)$$

then,

$$I_R = \frac{V_R}{R} \qquad (2)$$

A practical method to evaluate the current produced in an ionization chamber is by considering its relation with the radiation field. The current *Iin* obtained in an ionization chamber vented to air is related with the exposure rate $\bar{X}$ in mR/hr by the equation:

$$Iin = \left(\frac{V\ P\ \bar{X}}{2.35\,T} \right) 1.4 \times 10^{-14} \qquad [A] \qquad (3)$$

where:
V is the volume of the chamber in dm^3
P is the atmospheric pressure in mmHg.
T is the temperature in °K

b) Charge measurement.

An electrometer indirectly measures the charge by sensing a voltage drop across a capacitor (a typically value of 1000 pF) placed in the measuring circuit, as shown in Fig. 2.

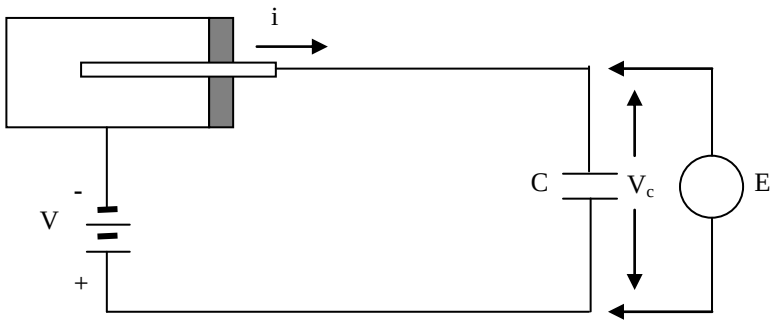


Fig. 2. Ionization charge measurement.

Where,
E = Electrometer.
C = Input capacitance of the electrometer.

i = Ionization current.

The voltage drop across the capacitor (C) is proportional to the integral input current in accordance with the formula:

$$V_c = \frac{1}{C}\int idt \qquad (4)$$

while the charge is obtained as:

$$q = \int idt \qquad (5)$$

then, the voltage is scaled and displayed as charge.

$$q = C.V_C \qquad (6)$$

3.3.2. Insulator and guard ring.

The electrodes of an ionization chamber are supported by a good insulator (see Fig. 3). If the ionization currents are extremely small in the center electrode, any leakage current (I_f) through this insulator can be added to the measured ionization current (I) and causes an unwanted component in the signal (I_T).

$$I_T = I_f + I \qquad (7)$$

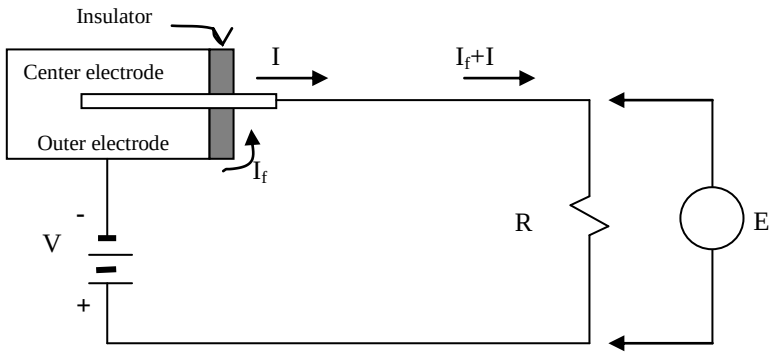


Fig. 3. Leakage current in an ionization chamber.

To eliminate this problem, the ionization chambers have a guard ring. As shown in Fig. 4. Always in low current applications of ionization chambers, a guard ring is employed to reduce the effects of insulator leakage. Normally a guard ring is manufactured with cylindrical geometry, then the insulator is divided into two parts, one of them is separated by the conducting guard ring from the negative electrode and the other part separating it from the positive electrode. Most of the voltage drop

occurs across the outer segment in which the resulting leakage current does not pass through the measuring electrometer (E)

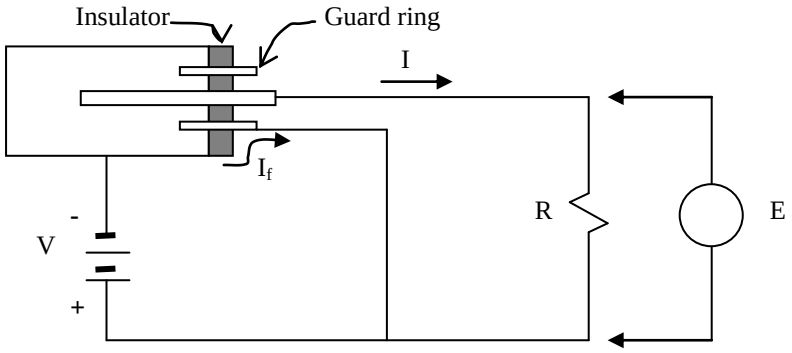


Fig. 4. Connection of an ionization chamber with guard ring.

3.3.3. Cables and connectors.

An ionization chamber needs to be connected to an electrometer through one cable with two connectors. Then it becomes an important part of the measuring process, because the connection cable is used to measure extremely small currents. Some characteristics of those cables and connectors are shown below.

a) Coaxial cable y triaxial cable.

A coaxial cable consists of a single conductor wire surrounded by a shield (see Fig. 5,a), while a traxial cable adds a second shield around the first one (see Fig. 5,b).
If someone wants to measure very small currents with those cables, both cables should be manufactured in low noise versions.

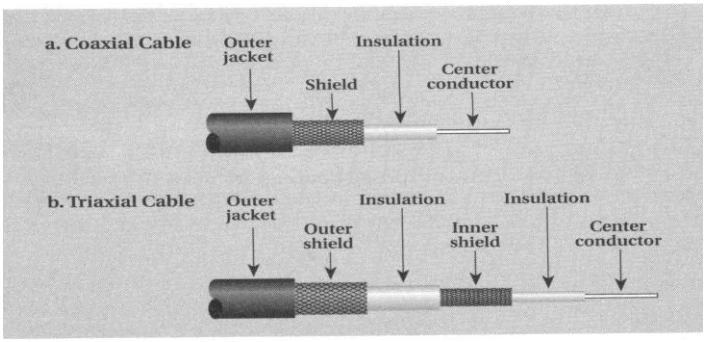


Fig. 5. Coaxial and triaxial cables.

The triaxial cable has an internal graphite coating to minimize current generated between the conductor wire and insulator due to friction (triboelectric effects).
The insulation resistance in these cables must be extremely high, mainly when it is used to measure high impedance, charge or current. For example, in a good quality

triaxial cable the insulators are made of polyethylene and have a typical insulation resistance of about 1 TΩ.

b) BNC connector.

The BNC connector is shown in Fig. 6, it includes a central conductor and shield. The central conductor of BNC connector is generally connected to measure the current, while the outer shell is usually connected to ground.

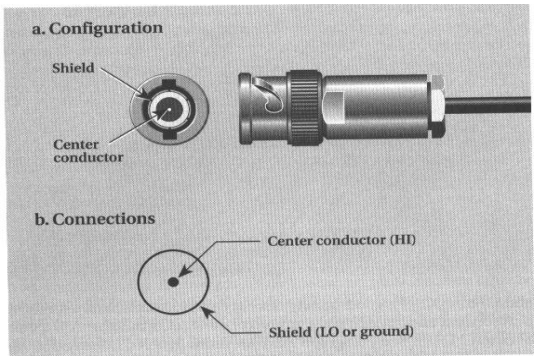


Fig. 6. BNC connector.

c) Triaxial connector.

The Triaxial connector is shown in Fig. 7, it includes a central conductor, inner shield and outer shield. The central conductor of the triaxial connector is generally connected to measure the current. The inner shield is connected to guard, while the outer shield is connected to chassis ground at the electrometer.

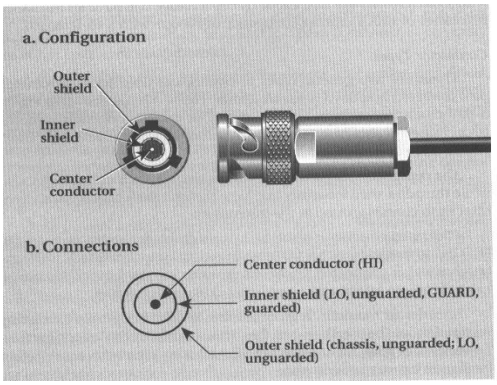


Fig. 7. Triaxial connector.

3.4. Previous operations.

Before beginning the test of the ionization chamber detector, make sure that the electrometer and connection cables are working correctly.

3.4.1 Cables and connectors.

Check the cables and connectors before starting the test because one fail in the connector or cable could give an incorrect result in low currents measurement.

i) The special cables and connectors shall be into a hermetic recipient with dry silica-gel, it can reduce the relative humidity and keeps clean the parts.

ii) Connectors.
Check the connectors which must be free of dust, fluff and metallic residuals that could increase the leakage current.

iii) Leakage current.
Check the ionization chamber with an electrometer, the leakage current must be not more than 15×10^{-15} A (15 fA).

3.4.2 Electrometer.

Check the calibration of the electrometer; it should be under the valid period.

3.5 Test of the leakage current without radiation.

The leakage current of an ionization chamber must be not more than 15 fA, because above this limit, the current measurement will not be correct. The leakage current is measured with an electrometer, the steps are shown below.

3.5.1 Connect the ionization chamber to electrometer, as shown in the figure of Annex II. Select the charge mode in the electrometer.

3.5.2 Enable the ZERO CHECK or ZERO CORRECT mode in the electrometer.

3.5.3 Disable ZERO CHECK and note the charge measurement (L_2) at the end of a specific interval of time (t). It is suggested to use 5 minutes.

3.5.4 Record the reading in the Table 1 of Annex IV.

3.5.5 To determine the leakage current (I_f), simply divide the measured charge by the time in seconds.

$$I_f = \frac{(L_2 - L_1)}{t} \qquad \frac{[C]}{[s]}, \ [A] \qquad (8)$$

Where,
L_1 = initial charge = 0 C

3.6. Test of the ionization chamber.

3.6.1 Voltage plateau for the ionization chamber.

A radioactive source with enough activity must be placed close to the ionization chamber in order to generate a current in the chamber.

The ionization chamber in Fig. 1 uses a voltage source. Increasing the applied voltage we could get a graphic of ionisation current as function of voltage (see Fig. 8).

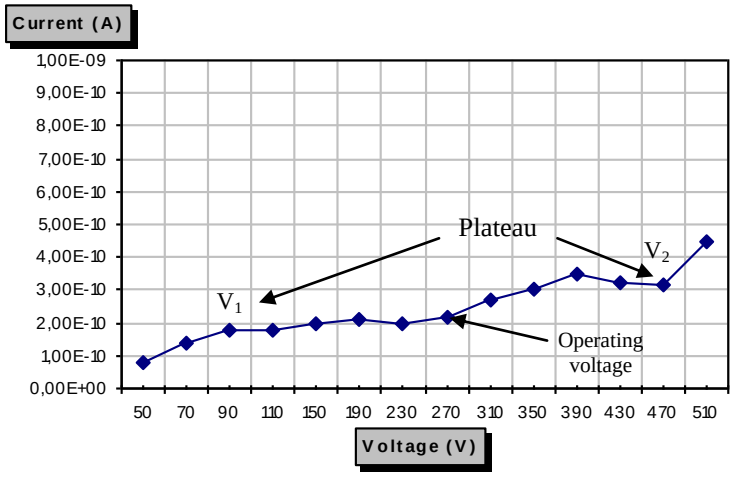


Fig. 8. Response of an ionization chamber as function of applied voltage.

Fig. 8 shows two parameters (voltage plateau and operating voltage) associated with the ionization current which is generated by an ionization chamber.

Connect the ionization chamber to electrometer and place the radioisotope source, as shown in Annex III. The geometry must be the same between the ionization chamber and the radioisotope source for each measurement. V_1 and I_1 , V_2 and I_2, are the readings of the high voltage applied and the electrometer current. Record these values in the Table 2 of Annex IV.

a) Voltage plateau or ionization chamber region.

An electric field can be crated by the aplicattion of a voltage inside the ionization chamber. If the field is sufficient to prevent the recombination of the original ion pairs, the detector reach the ideal operation mode for an ionization chamber, in this region the ionization current is constant, although the voltage increases.

The plateau lenght (L) is defined as:

$$L = V_1 - V_2 \tag{9}$$

In this example,

$$L = 480\,V - 90\,V = 390\,V$$

b) Operating voltage.

The operating voltage (V_O) is selected in the middle of the voltage plateau (see Fig. 8). Record this value in the Table 3 of Annex IV

$$V_O = V_1 + \frac{V_1 - V_2}{2} \qquad (10)$$

In this example,

$$Vo = 90\,V + \frac{480\,V - 90\,V}{2} = 285\,V$$

3.6.2 Sensitivity.

The sensitivity (S) to a radioactive source is obtained as the ratio between the ionization current (I) and the exposure rate ($\overline{X}$).

$$S = \frac{I[A]}{\overline{X}[R/hr]} \qquad (11)$$

The sensitivity as function of different exposure rates is illustrated in Fig. 9.

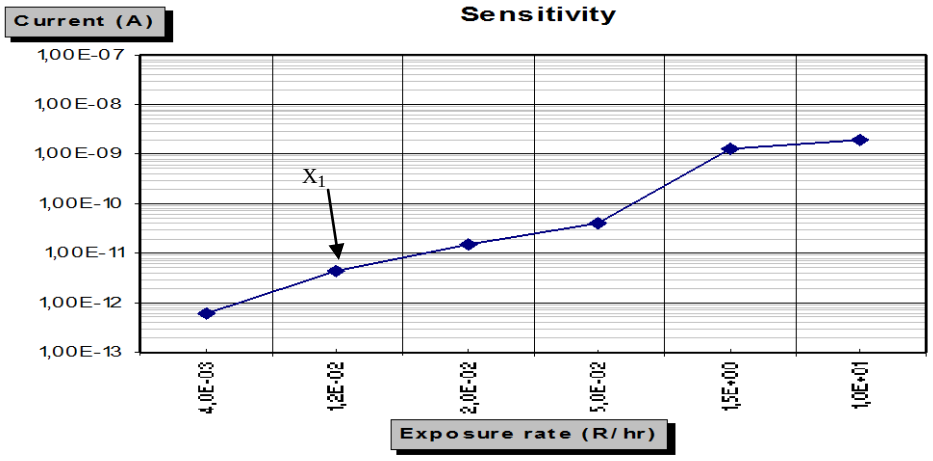


Fig. 9. Sensitivity curve for an ionization chamber with γ radiation.

Connect the ionization chamber to electrometer and place the radioisotope source, as shown in Annex III. The geometry is the same between the ionization chamber and

the radioisotope source for each measurement. $\overline{X}_1$ and I_1 , $\overline{X}_2$ and I_2 , etc. are several exposure rates and the readings of electrometer respectively. Record these values in the Table 4 of Annex IV.

The sensitivity in this example is calculated in one point in the first linear region. Record these values in the Table 5 of Annex IV.

$$S = \frac{I_1}{\overline{X}_1} \tag{12}$$

In this example

$$S = \frac{4.50 \times 10^{-12}[A]}{1.2 \times 10^{-2}[R/hr]} = 3.75 \times 10^{-10} \frac{A}{R/hr}$$

3.7 Ionization chamber aging.

Aging occurs and a detector changes its characteristics and response. Although, it can continue working but with some degradation. Some aging effects for an ionization chamber are described below:

- ✓ ***Leakage current***, increment of the leakage current. This effect can be produced by a degraded insulator, contaminants deposited on the surface of the insulator and high relative humidity in the environment.
- ✓ ***Low sensitivity***, decrement of the ionization current. This effect can be produced by graphite losses inside the chamber body and damage in the collector electrode.
- ✓ ***Short range of plateau***, in sealed chambers the quenching gas is gradually consumed through the time. This effect can reduce the lifetime of the chamber, its plateau length and operating voltage.

3.8 Test report.

3.8.1. The test report should include the following components: In the first part an descriptive title, identification number, information about the detector and information about the test equipments. The second part is the body of the technical report that includes: methods, results, graphics, conclusions and discussion, issue date, author name and reviewer name. Finally a list of references and separate appendices may also be included. The Annex IV shows an example.
The test report must be reviewed by the area responsible.

4.- ACTION IN CASE OF NON CONFORMITIES.

4.1 Technical Report.

Even in the case that results of the test are not as expected, a technical report has to be elaborated, indicating the non conformities and how far are the measured characteristics from the ideal ones.

4.2 Labelling.

The components or equipments that are not under specifications or with a failure have to be marked with a label indicating: **OUT OF SPECIFICATIONS** and **FAILURE** respectively.

5. RESPONSABILITIES.

5.1. Head of the Department.

The application of this procedure should be supervised by the head of the department.

5.2. Area responsible.

5.2.1. Assure that the test equipments are in good operational conditions and additionally are calibrated.
5.2.2. Supervise the activities for testing of ionization chambers.
5.2.3. Verify the test report.

5.3 Operative personnel.

5.3.1. Verify that the test equipments are in good operational conditions and additionally verify that all equipments are calibrated.
5.3.2. Apply the test procedure for ionization chambers detectors.
5.3.3 Elaborate the test report.

6. BIBLIOGRAPHY.

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3 Knoll, Glenn F. "RADIATION DETECTION AND MEASUREMENT", Third Edition, John Wiley and Sons. U.S.A. 2000.

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7 ANNEXES.

Annex I.
Flow chart.

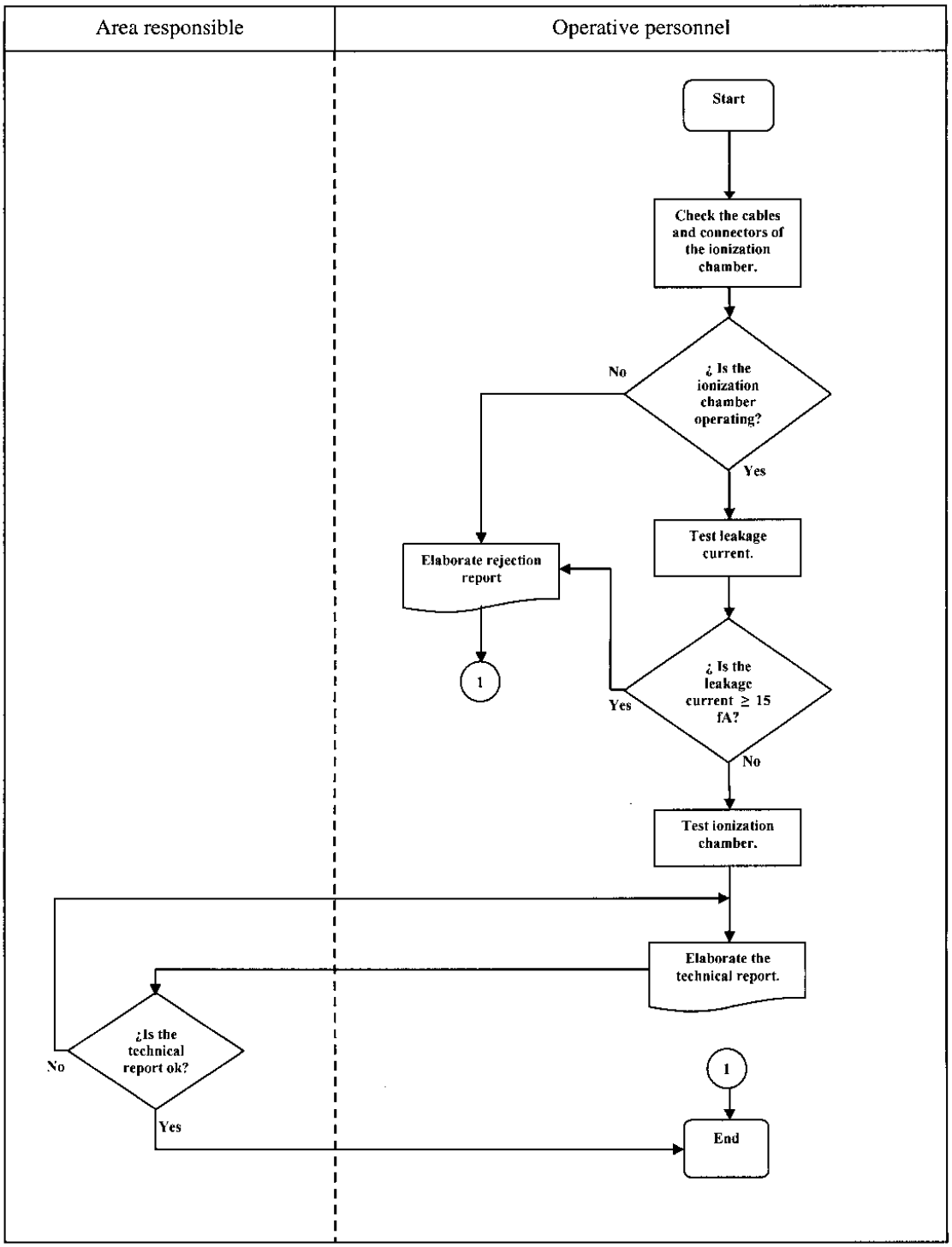
Annex II.
Connection between an electrometer and an ionization chamber.

Annex III.
Connection to test an ionization chamber with a radioactive source.

Annex IV.
Test report.

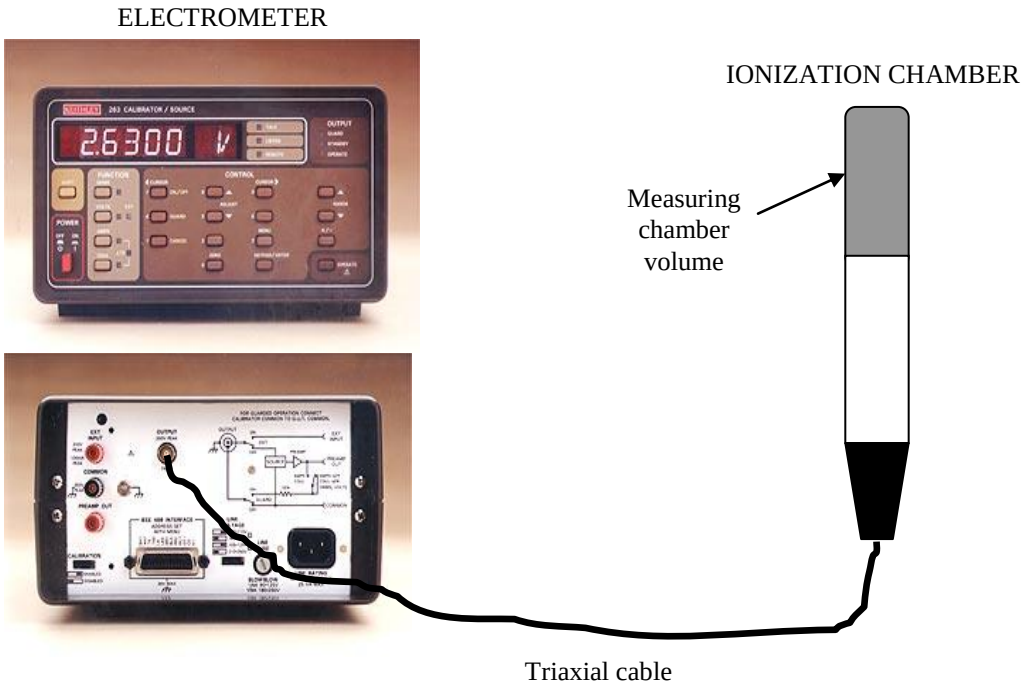
Annex I.

Flow chart.



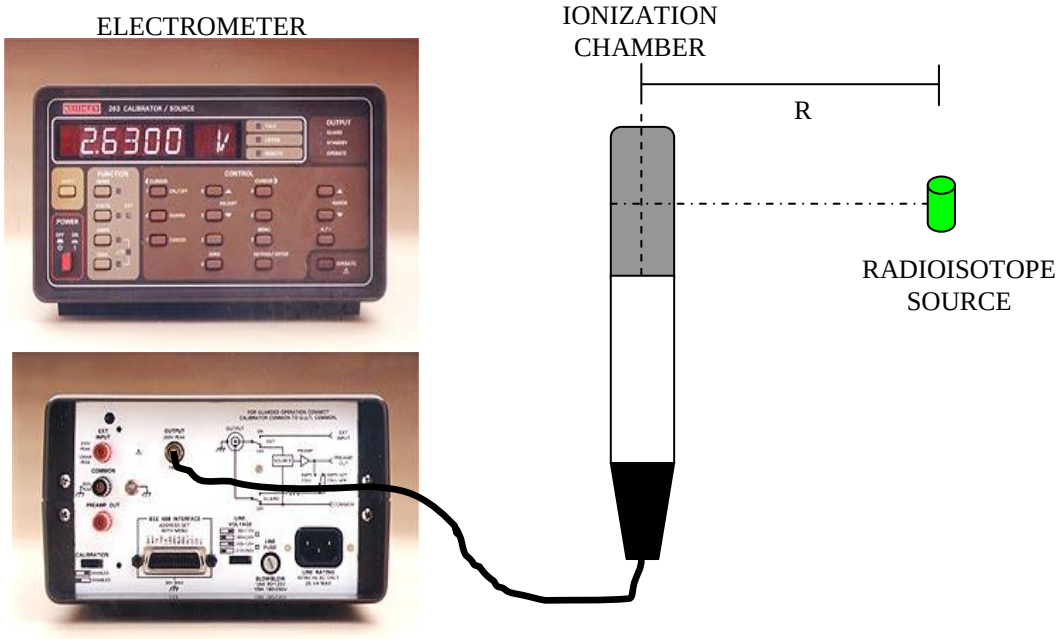
Annex II.

Connection between electrometer and an ionization chamber.



Annex III.

Connection to test an ionization chamber with a radioactive source.



Annex IV

TEST REPORT

TEST REPORT No. IC - _____

Ionization chamber

Manufacturer: _____ Model: _____

Serial number: _____ Window: _____

Test equipments

Manufacturer: _____ Model: _____

Serial number: _____

Manufacturer: _____ Model: _____

Serial number: _____

Radioactive source

Source: _____ Energy: _____

Activity: _____ Date: _____

Test Conditions

Temperature, ºK _____ Atmospheric Pressure, (mm Hg) _____

Relative Humidity, % _____

Results.

1. Test of the leakage current.

Table 1. Leakage current.

Charge	
Item	Measured value
L_1 =	
L_2 =	
Time	
	Proposed value
t =	
Leakage current	
	Calculated value
I_f =	

2. Voltage plateau.

Table 2. Readings.

Item	Voltage (V)	Current (A)
1		
2		
3		
4		
5		
6		
n		

n = number of readings are proposed by user

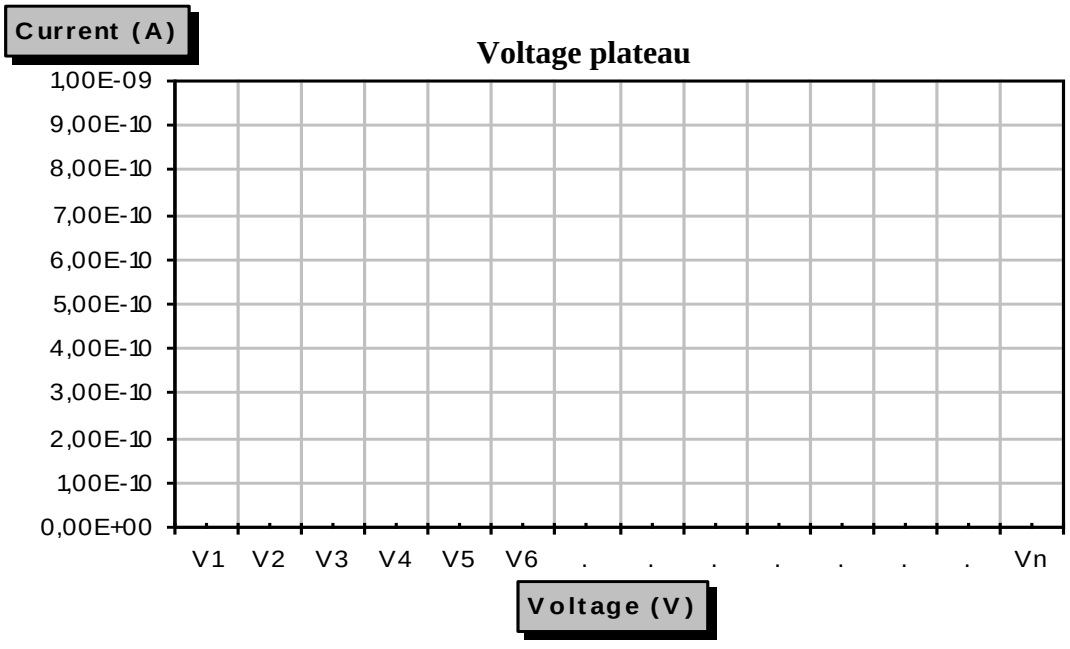


Table 3. Operating voltage.

Voltages	
Item	Measured value
V_1 =	
V_2 =	
Operating voltage	
	Calculated value
V_O =	

3. Test of sensitivity.

Table 4. Readings.

Item	Exposure rate (R/hr)	Current (A)
1		
2		
3		
4		
5		
6		
n		

n = number of readings are proposed by user

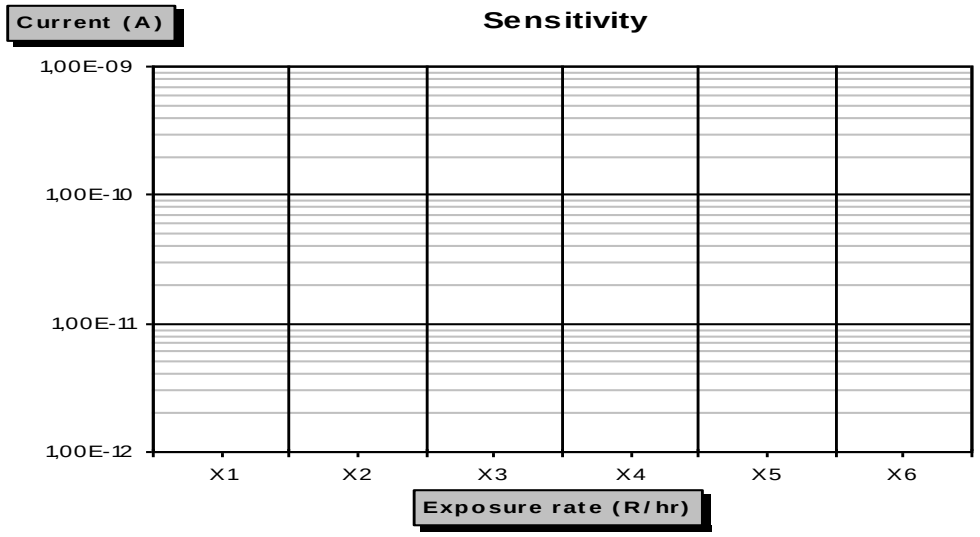


Table 5. Sensitivity.

Exposure rate	
Item	Measured value
$\overline{X}_1$ =	
Measured current	
Item	Measured value
I_1 =	
Sensitivity	
	Calculated value
S =	

4. Diagnostic or conclusions.

Tested by: _____

Reviewed by: _____

Date: _____